In [ ]:
```
%load_ext sql
```

In [ ]:
```
%sql mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
```

In [23]:
```
%%sql

select top(5) * from SuperStore_Data;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.
```

Out[23]:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida

EDA

1- Sales by City

2- Customers by Transactions

3- Sales by Segment

4- Customer Segmentation Based on Sales

5- Sales over Time (Years, Months, Days)

1- Sales by City

In [20]:
```
%%sql

Select top(5) City, Round(sum(Sales), 0) as Total_Sales
from SuperStore_Data
GROUP BY City
Order by Total_Sales desc;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.
```

Out[20]:

City	Total_Sales
New York City	256368.0
Los Angeles	175851.0
Seattle	119541.0
San Francisco	112669.0
Philadelphia	109077.0

2- Customers by Transactions

In [24]:
```
%%sql

Select top(5) Customer_Name, Count(Order_ID) as Tran_Num
from SuperStore_Data
Group By Customer_Name
Order By Tran_Num desc;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.
```

Out[24]:

Customer_Name	Tran_Num
William Brown	37
John Lee	34
Matt Abelman	34
Paul Prost	34
Seth Vernon	32

4- Customer Segmentation Based on Sales

In [46]:
```
%%sql

Create View Customer_Levels As

Select Customer_Name,Round(sum(sales),0) as Total_Sales,
  Case
    when sum(sales) > 10000 then 'VIP Customer'
    when sum(sales) > 5000 then 'AVG Customer'
    else 'Normal Customer'
End as Customer_Level

from SuperStore_Data
Group By Customer_Name;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
(pyodbc.ProgrammingError) ('42S01', "[42S01] [Microsoft][ODBC Driver 17 for SQL Server][SQL Server]There is already a
n object named 'Customer_Levels' in the database. (2714) (SQLExecDirectW)")
[SQL: Create View Customer_Levels As

Select Customer_Name,Round(sum(sales),0) as Total_Sales,
  Case
    when sum(sales) > 10000 then 'VIP Customer'
    when sum(sales) > 5000 then 'AVG Customer'
    else 'Normal Customer'
End as Customer_Level

from SuperStore_Data
Group By Customer_Name;]
(Background on this error at: https://sqlalche.me/e/20/f405)
```

In [47]:
```
%%sql

select * from Customer_Levels;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.
```

Out[47]:

Customer_Name	Total_Sales	Customer_Level
Michael Chen	3806.0	Normal Customer
Brian Moss	7294.0	AVG Customer
Tamara Chand	19052.0	VIP Customer
Justin MacKendrick	2834.0	Normal Customer
Doug Bickford	1989.0	Normal Customer
Jesus Ocampo	1091.0	Normal Customer
Sheri Gordon	1885.0	Normal Customer
John Huston	529.0	Normal Customer
Sean Miller	25043.0	VIP Customer
Liz Carlisle	2095.0	Normal Customer
Benjamin Patterson	1181.0	Normal Customer
Shahid Hopkins	2181.0	Normal Customer
Tracy Collins	743.0	Normal Customer
Eugene Moren	4588.0	Normal Customer
Alice McCarthy	814.0	Normal Customer
Natalie Fritzler	8323.0	AVG Customer
Arthur Prichep	3324.0	Normal Customer
Tracy Poddar	1937.0	Normal Customer
Ruben Ausman	3832.0	Normal Customer
Craig Yedwab	2900.0	Normal Customer
Thea Hendricks	2423.0	Normal Customer
Tim Brockman	1602.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Anthony Johnson	4501.0	Normal Customer
Noah Childs	1822.0	Normal Customer
Adam Bellavance	7756.0	AVG Customer
Scot Wooten	1085.0	Normal Customer
John Murray	7625.0	AVG Customer
Darren Koutras	1687.0	Normal Customer
Jill Matthias	304.0	Normal Customer
Erica Smith	3510.0	Normal Customer
Mary O'Rourke	922.0	Normal Customer
Phillip Flathmann	2206.0	Normal Customer
Ann Chong	2538.0	Normal Customer
Dave Kipp	2651.0	Normal Customer
Sean O'Donnell	2603.0	Normal Customer
Liz MacKendrick	1347.0	Normal Customer
Mark Cousins	4432.0	Normal Customer
Cathy Hwang	1537.0	Normal Customer
George Ashbrook	3920.0	Normal Customer
Andrew Allen	1791.0	Normal Customer
Logan Currie	2634.0	Normal Customer
Kristen Hastings	5991.0	AVG Customer
Maribeth Schnelling	7444.0	AVG Customer
Scot Coram	4372.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Ricardo Emerson	48.0	Normal Customer
Lena Radford	5143.0	AVG Customer
Mitch Willingham	5254.0	AVG Customer
Hilary Holden	219.0	Normal Customer
Russell D'Ascenzo	971.0	Normal Customer
Sanjit Chand	14142.0	VIP Customer
Magdelene Morse	1314.0	Normal Customer
Harry Greene	786.0	Normal Customer
Tanja Norvell	3130.0	Normal Customer
Gary Hansen	2819.0	Normal Customer
Aaron Bergman	886.0	Normal Customer
Pete Kriz	8647.0	AVG Customer
Ken Heidel	3919.0	Normal Customer
Jane Waco	7722.0	AVG Customer
Sarah Brown	6411.0	AVG Customer
Anemone Ratner	88.0	Normal Customer
Carlos Daly	2034.0	Normal Customer
Christopher Schild	3652.0	Normal Customer
Philip Fox	2770.0	Normal Customer
Deirdre Greer	3196.0	Normal Customer
Adrian Shami	59.0	Normal Customer
Justin Deggeller	8828.0	AVG Customer

Customer_Name	Total_Sales	Customer_Level
Mathew Reese	639.0	Normal Customer
George Bell	1568.0	Normal Customer
Mick Crebagga	2219.0	Normal Customer
Dorothy Badders	3909.0	Normal Customer
Lauren Leatherbury	2741.0	Normal Customer
Nicole Fjeld	2031.0	Normal Customer
Roger Demir	1420.0	Normal Customer
Brendan Sweed	2455.0	Normal Customer
Katherine Hughes	3101.0	Normal Customer
Patrick Ryan	1840.0	Normal Customer
Hunter Lopez	12873.0	VIP Customer
Tracy Zic	1707.0	Normal Customer
Penelope Sewall	6844.0	AVG Customer
Brad Eason	1728.0	Normal Customer
Ross DeVincentis	1138.0	Normal Customer
Luke Schmidt	1010.0	Normal Customer
Henia Zydlo	887.0	Normal Customer
Dana Kaydos	4282.0	Normal Customer
Benjamin Farhat	1585.0	Normal Customer
Carl Weiss	1137.0	Normal Customer
Anne Pryor	1639.0	Normal Customer
Christina VanderZanden	2831.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Tom Ashbrook	14596.0	VIP Customer
Harold Dahlen	251.0	Normal Customer
Ann Steele	833.0	Normal Customer
Ryan Akin	1687.0	Normal Customer
Liz Preis	3653.0	Normal Customer
Patrick O'Donnell	2493.0	Normal Customer
Hallie Redmond	1299.0	Normal Customer
Trudy Brown	2798.0	Normal Customer
Steve Carroll	1255.0	Normal Customer
Mitch Gastineau	17.0	Normal Customer
Keith Herrera	5953.0	AVG Customer
Becky Pak	1698.0	Normal Customer
Erin Mull	229.0	Normal Customer
Chuck Sachs	551.0	Normal Customer
Aaron Smayling	3051.0	Normal Customer
Shahid Shariari	3057.0	Normal Customer
Michael Paige	3984.0	Normal Customer
John Grady	1507.0	Normal Customer
Troy Staebel	2820.0	Normal Customer
Mike Vittorini	1735.0	Normal Customer
Alan Barnes	1114.0	Normal Customer
Seth Vernon	11471.0	VIP Customer

Customer_Name	Total_Sales	Customer_Level
Rick Hansen	3621.0	Normal Customer
Shui Tom	433.0	Normal Customer
Max Ludwig	2072.0	Normal Customer
Art Ferguson	2053.0	Normal Customer
Sarah Bern	1348.0	Normal Customer
Jeremy Pistek	3636.0	Normal Customer
John Lee	9800.0	AVG Customer
Natalie Webber	7234.0	AVG Customer
Toby Braunhardt	2198.0	Normal Customer
Evan Henry	924.0	Normal Customer
Alan Shonely	585.0	Normal Customer
David Flashing	2391.0	Normal Customer
Tracy Hopkins	4234.0	Normal Customer
Emily Burns	2767.0	Normal Customer
Denny Ordway	1300.0	Normal Customer
Marc Crier	2726.0	Normal Customer
Heather Kirkland	4878.0	Normal Customer
Ann Blume	1516.0	Normal Customer
Maria Etezadi	10664.0	VIP Customer
Valerie Takahito	1737.0	Normal Customer
Rose O'Brian	3815.0	Normal Customer
Barbara Fisher	600.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Carol Triggs	3242.0	Normal Customer
Nancy Lomonaco	1418.0	Normal Customer
Brad Norvell	265.0	Normal Customer
Catherine Glotzbach	904.0	Normal Customer
Henry Goldwyn	3248.0	Normal Customer
Amy Cox	5528.0	AVG Customer
Patrick Jones	1220.0	Normal Customer
Anthony Garverick	171.0	Normal Customer
Barry Blumstein	744.0	Normal Customer
Julia Dunbar	695.0	Normal Customer
Scott Williamson	1967.0	Normal Customer
Craig Carroll	2854.0	Normal Customer
Hunter Glantz	3690.0	Normal Customer
Alyssa Crouse	926.0	Normal Customer
Naresj Patel	5530.0	AVG Customer
Don Miller	1377.0	Normal Customer
Arthur Gainer	4511.0	Normal Customer
Jay Fein	1912.0	Normal Customer
Clay Rozendal	2149.0	Normal Customer
Brendan Murry	808.0	Normal Customer
Anna Andreadi	5087.0	AVG Customer
Susan Gilcrest	48.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Ken Black	2745.0	Normal Customer
Ed Braxton	3645.0	Normal Customer
Edward Hooks	10311.0	VIP Customer
Tony Chapman	1245.0	Normal Customer
Julie Creighton	859.0	Normal Customer
Charlotte Melton	1475.0	Normal Customer
Filia McAdams	2457.0	Normal Customer
Michelle Tran	1319.0	Normal Customer
Matt Connell	2258.0	Normal Customer
Jeremy Farry	1083.0	Normal Customer
Max Jones	6321.0	AVG Customer
Maribeth Dona	3766.0	Normal Customer
Jamie Frazer	2280.0	Normal Customer
Cindy Stewart	5690.0	AVG Customer
Rick Huthwaite	6979.0	AVG Customer
Christine Abelman	1422.0	Normal Customer
Ralph Kennedy	1460.0	Normal Customer
Greg Tran	11820.0	VIP Customer
Dave Poirier	2215.0	Normal Customer
Larry Blacks	50.0	Normal Customer
Giulietta Weimer	2189.0	Normal Customer
Darrin Van Huff	1119.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Duane Noonan	2140.0	Normal Customer
Ashley Jarboe	1839.0	Normal Customer
Alex Avila	5564.0	AVG Customer
Michael Moore	3794.0	Normal Customer
Anthony Witt	649.0	Normal Customer
Rick Reed	3302.0	Normal Customer
Neil Knudson	572.0	Normal Customer
Sung Pak	4283.0	Normal Customer
Greg Hansen	147.0	Normal Customer
Fred Chung	3889.0	Normal Customer
Toby Ritter	676.0	Normal Customer
Joy Daniels	385.0	Normal Customer
Grace Kelly	6497.0	AVG Customer
Trudy Glocke	2075.0	Normal Customer
Grant Thornton	9351.0	AVG Customer
Richard Eichhorn	877.0	Normal Customer
Julia West	1429.0	Normal Customer
Dianna Arnett	157.0	Normal Customer
Karen Ferguson	10604.0	VIP Customer
Cindy Schnelling	1077.0	Normal Customer
Ryan Crowe	886.0	Normal Customer
Jennifer Braxton	1792.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Andy Gerbode	1455.0	Normal Customer
Tom Prescott	5329.0	AVG Customer
Cassandra Brandow	6076.0	AVG Customer
Karen Carlisle	2121.0	Normal Customer
John Stevenson	3868.0	Normal Customer
Anne McFarland	5664.0	AVG Customer
Dennis Kane	3318.0	Normal Customer
Jim Sink	1131.0	Normal Customer
Eric Barreto	945.0	Normal Customer
David Wiener	1081.0	Normal Customer
Anna Chung	657.0	Normal Customer
Jack Lebron	1215.0	Normal Customer
Alejandro Savely	3214.0	Normal Customer
Luke Foster	3931.0	Normal Customer
Ivan Liston	5041.0	AVG Customer
Erin Ashbrook	2847.0	Normal Customer
Barry Französisch	2889.0	Normal Customer
Toby Swindell	975.0	Normal Customer
Stewart Visinsky	4105.0	Normal Customer
Chris Cortes	3913.0	Normal Customer
Eleni McCrary	2567.0	Normal Customer
Claire Gute	1149.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Zuschuss Donatelli	1494.0	Normal Customer
Charles Crestani	2472.0	Normal Customer
Liz Thompson	2936.0	Normal Customer
Pauline Johnson	1824.0	Normal Customer
Ivan Gibson	745.0	Normal Customer
Raymond Messe	1453.0	Normal Customer
Katrina Bavinger	909.0	Normal Customer
Jeremy Ellison	1389.0	Normal Customer
Ellis Ballard	4358.0	Normal Customer
Laurel Elliston	1469.0	Normal Customer
Vivian Mathis	381.0	Normal Customer
Claudia Bergmann	4246.0	Normal Customer
Mike Gockenbach	3062.0	Normal Customer
Nat Carroll	2461.0	Normal Customer
Michelle Moray	2750.0	Normal Customer
Bradley Nguyen	1459.0	Normal Customer
Dennis Pardue	5481.0	AVG Customer
Bill Stewart	3888.0	Normal Customer
Corey Catlett	2541.0	Normal Customer
Mitch Webber	5342.0	AVG Customer
Beth Fritzler	792.0	Normal Customer
Stephanie Ulpright	2595.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Alan Schoenberger	4261.0	Normal Customer
Denise Leinenbach	585.0	Normal Customer
Karen Seio	88.0	Normal Customer
Joseph Holt	7955.0	AVG Customer
Helen Wasserman	9300.0	AVG Customer
Thomas Brumley	816.0	Normal Customer
Brendan Dodson	321.0	Normal Customer
Maureen Gastineau	2350.0	Normal Customer
Don Jones	2502.0	Normal Customer
Bradley Drucker	4411.0	Normal Customer
Gary McGarr	3583.0	Normal Customer
Bill Eplett	4205.0	Normal Customer
Brian DeCherney	1538.0	Normal Customer
Cathy Armstrong	1680.0	Normal Customer
Harry Marie	8237.0	AVG Customer
Corinna Mitchell	6340.0	AVG Customer
Juliana Krohn	3748.0	Normal Customer
Steve Nguyen	3128.0	Normal Customer
Jeremy Lonsdale	3174.0	Normal Customer
Paul Knutson	1441.0	Normal Customer
Karen Bern	3153.0	Normal Customer
Roland Fjeld	2341.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Todd Boyes	834.0	Normal Customer
Julie Kriz	2932.0	Normal Customer
Maria Zettner	594.0	Normal Customer
Roger Barcio	2067.0	Normal Customer
Raymond Buch	15117.0	VIP Customer
Guy Armstrong	1398.0	Normal Customer
Robert Waldorf	3496.0	Normal Customer
Gary Hwang	4173.0	Normal Customer
Randy Bradley	1008.0	Normal Customer
Kunst Miller	4909.0	Normal Customer
Resi Pölking	5153.0	AVG Customer
Chris Selesnick	2754.0	Normal Customer
Chris McAfee	2306.0	Normal Customer
Julie Prescott	1708.0	Normal Customer
Katrina Willman	4417.0	Normal Customer
Bruce Geld	1006.0	Normal Customer
Darren Budd	2839.0	Normal Customer
Nick Zandusky	2488.0	Normal Customer
Carol Darley	640.0	Normal Customer
Jennifer Halladay	309.0	Normal Customer
Dorris liebe	756.0	Normal Customer
Neil Französisch	377.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Phillina Ober	1057.0	Normal Customer
Matthew Grinstein	3785.0	Normal Customer
Frank Hawley	4256.0	Normal Customer
Kelly Andreada	1520.0	Normal Customer
Erin Creighton	462.0	Normal Customer
Delfina Latchford	1458.0	Normal Customer
Alyssa Tate	1172.0	Normal Customer
Lori Olson	644.0	Normal Customer
Joy Bell-	644.0	Normal Customer
Sarah Foster	3423.0	Normal Customer
Max Engle	1908.0	Normal Customer
Larry Hughes	287.0	Normal Customer
Pete Takahito	161.0	Normal Customer
Steven Cartwright	5226.0	AVG Customer
Allen Armold	1056.0	Normal Customer
James Galang	6366.0	AVG Customer
Janet Molinari	1804.0	Normal Customer
Sanjit Jacobs	3950.0	Normal Customer
Marina Lichtenstein	2723.0	Normal Customer
Adrian Hane	1736.0	Normal Customer
Becky Castell	2934.0	Normal Customer
Christopher Conant	12129.0	VIP Customer

Customer_Name	Total_Sales	Customer_Level
Amy Hunt	2495.0	Normal Customer
Erica Hernandez	1220.0	Normal Customer
Mike Kennedy	1032.0	Normal Customer
Katrina Edelman	1687.0	Normal Customer
Barry Pond	983.0	Normal Customer
Tracy Blumstein	4737.0	Normal Customer
Shaun Weien	2922.0	Normal Customer
Valerie Dominguez	6442.0	AVG Customer
Gene McClure	1256.0	Normal Customer
Tonja Turnell	5365.0	AVG Customer
Chad McGuire	1662.0	Normal Customer
Susan Pistek	3991.0	Normal Customer
Anthony O'Donnell	161.0	Normal Customer
Dean Katz	4802.0	Normal Customer
Jessica Myrick	1733.0	Normal Customer
Michael Nguyen	2478.0	Normal Customer
Irene Maddox	4930.0	Normal Customer
Roy Französisch	945.0	Normal Customer
Darrin Martin	6345.0	AVG Customer
Greg Guthrie	2224.0	Normal Customer
Denny Joy	2013.0	Normal Customer
Joni Wasserman	1244.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Lisa Ryan	1879.0	Normal Customer
Clay Ludtke	10881.0	VIP Customer
Maribeth Yedwab	2254.0	Normal Customer
Jocasta Rupert	864.0	Normal Customer
Nora Pelletier	1229.0	Normal Customer
Maxwell Schwartz	3145.0	Normal Customer
Peter Bühler	688.0	Normal Customer
Yana Sorensen	6720.0	AVG Customer
Caroline Jumper	11165.0	VIP Customer
Christina Anderson	2648.0	Normal Customer
Nick Radford	2367.0	Normal Customer
Theone Pippenger	4454.0	Normal Customer
Daniel Raglin	8351.0	AVG Customer
Nona Balk	1973.0	Normal Customer
Dave Brooks	4532.0	Normal Customer
Rob Beeghly	529.0	Normal Customer
Nicole Hansen	2673.0	Normal Customer
Kean Thornton	4642.0	Normal Customer
Eva Jacobs	2657.0	Normal Customer
Joe Elijah	8698.0	AVG Customer
Fred Wasserman	80.0	Normal Customer
Maris LaWare	2921.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Michelle Huthwaite	2985.0	Normal Customer
Becky Martin	11790.0	VIP Customer
Brad Thomas	415.0	Normal Customer
Jason Fortune-	955.0	Normal Customer
Vicky Freymann	848.0	Normal Customer
Lycoris Saunders	369.0	Normal Customer
Jay Kimmel	2710.0	Normal Customer
Bill Tyler	2187.0	Normal Customer
Matthew Clasen	795.0	Normal Customer
Lisa Hazard	318.0	Normal Customer
Sample Company A	4625.0	Normal Customer
Suzanne McNair	5563.0	AVG Customer
Adrian Barton	14474.0	VIP Customer
Karen Daniels	8282.0	AVG Customer
Clytie Kelty	2373.0	Normal Customer
Chuck Clark	2870.0	Normal Customer
Meg Tillman	2890.0	Normal Customer
Matt Collins	1304.0	Normal Customer
Carlos Meador	565.0	Normal Customer
Gene Hale	1361.0	Normal Customer
Robert Dilbeck	2787.0	Normal Customer
Neoma Murray	2162.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Guy Phonely	237.0	Normal Customer
Natalie DeCherney	4326.0	Normal Customer
Monica Federle	1758.0	Normal Customer
Brosina Hoffman	6255.0	AVG Customer
Clay Cheatham	114.0	Normal Customer
Carlos Soltero	3037.0	Normal Customer
Giulietta Dortch	1888.0	Normal Customer
Benjamin Venier	1523.0	Normal Customer
Dionis Lloyd	3539.0	Normal Customer
Barry Gonzalez	2799.0	Normal Customer
Jim Kriz	4760.0	Normal Customer
Jenna Caffey	1058.0	Normal Customer
Lena Creighton	7663.0	AVG Customer
Olvera Toch	3819.0	Normal Customer
Pamela Coakley	1832.0	Normal Customer
Quincy Jones	6108.0	AVG Customer
Arthur Wiediger	2853.0	Normal Customer
Eric Murdock	934.0	Normal Customer
John Lucas	2725.0	Normal Customer
Mark Hamilton	2456.0	Normal Customer
Ionia McGrath	2873.0	Normal Customer
Tamara Dahlen	1435.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Berenike Kampe	659.0	Normal Customer
Alejandro Ballentine	915.0	Normal Customer
Astrea Jones	629.0	Normal Customer
Adam Hart	3250.0	Normal Customer
Kalyca Meade	2261.0	Normal Customer
Larry Tron	330.0	Normal Customer
Maureen Gnade	1342.0	Normal Customer
Allen Rosenblatt	2236.0	Normal Customer
Mick Brown	1428.0	Normal Customer
Vivek Sundaresam	1056.0	Normal Customer
Shirley Schmidt	613.0	Normal Customer
Toby Gnade	3058.0	Normal Customer
Nathan Gelder	1396.0	Normal Customer
Nora Preis	7903.0	AVG Customer
Keith Dawkins	8181.0	AVG Customer
John Castell	2772.0	Normal Customer
Sean Wendt	1019.0	Normal Customer
Bill Shonely	10502.0	VIP Customer
Frank Carlisle	3419.0	Normal Customer
Matt Abelman	4299.0	Normal Customer
Sean Christensen	3979.0	Normal Customer
James Lanier	1572.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
David Kendrick	798.0	Normal Customer
Katherine Nockton	2617.0	Normal Customer
Pierre Wener	3922.0	Normal Customer
Barry Weirich	386.0	Normal Customer
Fred Hopkins	6987.0	AVG Customer
Anthony Rawles	4523.0	Normal Customer
Katharine Harms	3313.0	Normal Customer
Ben Wallace	1738.0	Normal Customer
Jennifer Ferguson	2371.0	Normal Customer
Rachel Payne	955.0	Normal Customer
Randy Ferguson	3309.0	Normal Customer
Lindsay Castell	3247.0	Normal Customer
Frank Preis	4047.0	Normal Customer
Paul Van Hugh	2876.0	Normal Customer
Frank Gastineau	1383.0	Normal Customer
Ed Jacobs	2162.0	Normal Customer
Rick Bensley	4715.0	Normal Customer
Sean Braxton	8058.0	AVG Customer
Muhammed MacIntyre	775.0	Normal Customer
Joe Kamberova	4867.0	Normal Customer
Brooke Gillingham	1874.0	Normal Customer
Erica Hackney	826.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Saphhira Shifley	1324.0	Normal Customer
Karl Braun	1569.0	Normal Customer
Tamara Manning	2955.0	Normal Customer
Bart Watters	4750.0	Normal Customer
Carl Ludwig	1262.0	Normal Customer
Michelle Arnett	2453.0	Normal Customer
Odella Nelson	1568.0	Normal Customer
Lindsay Williams	3230.0	Normal Customer
Michael Oakman	154.0	Normal Customer
Steve Chapman	2576.0	Normal Customer
Charles McCrossin	2533.0	Normal Customer
Kelly Williams	449.0	Normal Customer
Paul Gonzalez	1987.0	Normal Customer
Dan Lawera	1503.0	Normal Customer
Peter McVee	4116.0	Normal Customer
Doug O'Connell	1267.0	Normal Customer
Roy Collins	966.0	Normal Customer
Steven Ward	759.0	Normal Customer
Mark Van Huff	1560.0	Normal Customer
Ed Ludwig	124.0	Normal Customer
Matt Collister	2426.0	Normal Customer
Sonia Cooley	903.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Jim Karlsson	3760.0	Normal Customer
Shahid Collister	5993.0	AVG Customer
Angele Hood	739.0	Normal Customer
Thomas Boland	3589.0	Normal Customer
Ken Brennan	984.0	Normal Customer
Alan Hwang	4805.0	Normal Customer
Nathan Mautz	6459.0	AVG Customer
Chloris Kastensmidt	3155.0	Normal Customer
Jim Epp	7755.0	AVG Customer
Luke Weiss	5421.0	AVG Customer
Lena Cacioppo	2581.0	Normal Customer
Logan Haushalter	1740.0	Normal Customer
Nick Crebassa	8242.0	AVG Customer
Christine Kargatis	2945.0	Normal Customer
Corey-Lock	1300.0	Normal Customer
Arianne Irving	4376.0	Normal Customer
Michelle Ellison	1257.0	Normal Customer
Greg Maxwell	6050.0	AVG Customer
Theresa Coyne	1038.0	Normal Customer
Christine Phan	5888.0	AVG Customer
Jennifer Jackson	709.0	Normal Customer
Scott Cohen	3730.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Aleksandra Gannaway	368.0	Normal Customer
Valerie Mitchum	2455.0	Normal Customer
Dean percer	7199.0	AVG Customer
Nat Gilpin	1870.0	Normal Customer
Paul MacIntyre	1405.0	Normal Customer
Stephanie Phelps	1872.0	Normal Customer
Joel Jenkins	195.0	Normal Customer
Edward Becker	3236.0	Normal Customer
Dorothy Dickinson	270.0	Normal Customer
Harold Engle	884.0	Normal Customer
Philip Brown	1457.0	Normal Customer
Bruce Degenhardt	1527.0	Normal Customer
Muhammed Lee	162.0	Normal Customer
Frank Olsen	2678.0	Normal Customer
Greg Matthias	3164.0	Normal Customer
Herbert Flentye	2533.0	Normal Customer
Shirley Jackson	1319.0	Normal Customer
Dan Campbell	3336.0	Normal Customer
Jason Klamczynski	384.0	Normal Customer
Muhammed Yedwab	4153.0	Normal Customer
Toby Carlisle	4267.0	Normal Customer
Mark Haberlin	400.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Joni Blumstein	901.0	Normal Customer
Anthony Jacobs	4867.0	Normal Customer
Jill Stevenson	1418.0	Normal Customer
Brian Thompson	3197.0	Normal Customer
Charles Sheldon	844.0	Normal Customer
Chad Sievert	885.0	Normal Customer
Kimberly Carter	1146.0	Normal Customer
Troy Blackwell	1388.0	Normal Customer
Mike Caudle	1582.0	Normal Customer
Michael Dominguez	1205.0	Normal Customer
Bill Overfelt	2683.0	Normal Customer
Cindy Chapman	1659.0	Normal Customer
Cyra Reiten	1398.0	Normal Customer
Debra Catini	1175.0	Normal Customer
Aimee Bixby	967.0	Normal Customer
Carl Jackson	17.0	Normal Customer
Victoria Brennan	2006.0	Normal Customer
Roy Phan	2171.0	Normal Customer
Sandra Flanagan	764.0	Normal Customer
Victoria Wilson	6134.0	AVG Customer
Doug Jacobs	3356.0	Normal Customer
Roland Schwarz	4160.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Philisse Overcash	4893.0	Normal Customer
Liz Pelletier	757.0	Normal Customer
Pauline Chand	1061.0	Normal Customer
Patrick Bzostek	740.0	Normal Customer
Pete Armstrong	729.0	Normal Customer
Steven Roelle	4346.0	Normal Customer
Christy Brittain	1949.0	Normal Customer
Brian Stugart	3288.0	Normal Customer
Matt Hagelstein	1253.0	Normal Customer
Ricardo Sperren	3221.0	Normal Customer
Susan Vittorini	1903.0	Normal Customer
Eugene Barchas	3071.0	Normal Customer
Guy Thornton	226.0	Normal Customer
Art Foster	862.0	Normal Customer
Bradley Talbott	2684.0	Normal Customer
Daniel Byrd	2172.0	Normal Customer
Tony Molinari	1095.0	Normal Customer
Beth Paige	2475.0	Normal Customer
Barry Franz	1334.0	Normal Customer
Bobby Elias	2261.0	Normal Customer
Denise Monton	4074.0	Normal Customer
Dave Hallsten	4933.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Michael Stewart	855.0	Normal Customer
Mike Pelletier	5088.0	AVG Customer
Julia Barnett	2518.0	Normal Customer
David Smith	2882.0	Normal Customer
Sarah Jordon	1341.0	Normal Customer
Roy Skaria	22.0	Normal Customer
Brian Dahlen	3288.0	Normal Customer
Justin Ritter	2202.0	Normal Customer
Skye Norling	1860.0	Normal Customer
Evan Bailliet	1186.0	Normal Customer
Chuck Magee	288.0	Normal Customer
Thomas Thornton	1427.0	Normal Customer
Lynn Smith	1008.0	Normal Customer
Meg O'Connel	1687.0	Normal Customer
Stefania Perrino	5440.0	AVG Customer
Andy Reiter	6608.0	AVG Customer
Bobby Trafton	604.0	Normal Customer
Harold Pawlan	1990.0	Normal Customer
Lindsay Shagiari	2989.0	Normal Customer
Dianna Vittorini	3342.0	Normal Customer
Christine Sundaresam	3202.0	Normal Customer
Sam Craven	3363.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Alex Russell	1056.0	Normal Customer
Patrick O'Brill	7474.0	AVG Customer
Rob Lucas	6235.0	AVG Customer
Craig Reiter	1601.0	Normal Customer
MaryBeth Skach	476.0	Normal Customer
Stewart Carmichael	4493.0	Normal Customer
Sandra Glassco	3407.0	Normal Customer
Zuschuss Carroll	8026.0	AVG Customer
Eugene Hildebrand	3083.0	Normal Customer
Helen Abelman	2163.0	Normal Customer
Brian Derr	583.0	Normal Customer
Jasper Cacioppo	71.0	Normal Customer
Sibella Parks	1306.0	Normal Customer
Beth Thompson	2568.0	Normal Customer
Stuart Calhoun	279.0	Normal Customer
Joni Sundaresam	469.0	Normal Customer
Allen Goldenen	201.0	Normal Customer
Phillip Breyer	133.0	Normal Customer
Deanra Eno	2551.0	Normal Customer
Pamela Stobb	2181.0	Normal Customer
Linda Cazamias	4493.0	Normal Customer
Jonathan Howell	959.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Dianna Wilson	5272.0	AVG Customer
Craig Carreira	2270.0	Normal Customer
Corey Roper	476.0	Normal Customer
Rick Wilson	7397.0	AVG Customer
Jack Garza	2359.0	Normal Customer
Aaron Hawkins	1745.0	Normal Customer
Deborah Brumfield	4656.0	Normal Customer
Fred Harton	2271.0	Normal Customer
Darren Powers	1051.0	Normal Customer
Helen Andreada	2584.0	Normal Customer
Frank Merwin	3736.0	Normal Customer
Darrin Sayre	1762.0	Normal Customer
Alejandro Grove	2583.0	Normal Customer
Joel Eaton	6761.0	AVG Customer
Richard Bierner	2663.0	Normal Customer
Eileen Kiefer	1303.0	Normal Customer
Cyma Kinney	1582.0	Normal Customer
Elizabeth Moffitt	2340.0	Normal Customer
Ralph Ritter	616.0	Normal Customer
Nathan Cano	2219.0	Normal Customer
Thomas Seio	5371.0	AVG Customer
Frank Atkinson	2984.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Sung Shariari	965.0	Normal Customer
Shaun Chance	1875.0	Normal Customer
Jamie Kunitz	4427.0	Normal Customer
Ritsa Hightower	740.0	Normal Customer
Jennifer Patt	1669.0	Normal Customer
Paul Stevenson	1279.0	Normal Customer
Cynthia Voltz	2370.0	Normal Customer
Kristina Nunn	2281.0	Normal Customer
Katherine Ducich	1889.0	Normal Customer
Xylona Preis	2375.0	Normal Customer
Sanjit Engle	12209.0	VIP Customer
Cynthia Delaney	1861.0	Normal Customer
Rob Williams	3280.0	Normal Customer
Melanie Seite	370.0	Normal Customer
George Zrebassa	2455.0	Normal Customer
Pauline Webber	3455.0	Normal Customer
Emily Ducich	3485.0	Normal Customer
Dario Medina	1322.0	Normal Customer
Jason Gross	2241.0	Normal Customer
Tom Stivers	2054.0	Normal Customer
Maya Herman	2204.0	Normal Customer
John Dryer	2152.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Janet Lee	2204.0	Normal Customer
Laurel Beltran	4986.0	Normal Customer
Janet Martin	3159.0	Normal Customer
Ted Trevino	4916.0	Normal Customer
Ben Ferrer	5908.0	AVG Customer
Linda Southworth	2845.0	Normal Customer
Duane Benoit	1155.0	Normal Customer
Michelle Lonsdale	742.0	Normal Customer
Sally Knutson	883.0	Normal Customer
Sue Ann Reed	4767.0	Normal Customer
Laura Armstrong	8673.0	AVG Customer
Laurel Workman	2238.0	Normal Customer
Neola Schneider	2576.0	Normal Customer
Anna Häberlin	7888.0	AVG Customer
Kelly Collister	3908.0	Normal Customer
Patrick Gardner	3087.0	Normal Customer
Damala Kotsonis	3256.0	Normal Customer
Robert Barroso	221.0	Normal Customer
Duane Huffman	1043.0	Normal Customer
Shirley Daniels	6121.0	AVG Customer
Katherine Murray	5620.0	AVG Customer
Denny Blanton	1712.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Annie Zypern	1622.0	Normal Customer
Ross Baird	5633.0	AVG Customer
Chad Cunningham	1771.0	Normal Customer
Rick Duston	4273.0	Normal Customer
Jill Fjeld	4198.0	Normal Customer
Maurice Satty	1613.0	Normal Customer
Paul Prost	7253.0	AVG Customer
Andrew Gjertsen	2357.0	Normal Customer
Sonia Sunley	418.0	Normal Customer
Nora Paige	3908.0	Normal Customer
Andrew Roberts	265.0	Normal Customer
Annie Thurman	3832.0	Normal Customer
Trudy Schmidt	3368.0	Normal Customer
Jonathan Doherty	7611.0	AVG Customer
Bart Pistole	2442.0	Normal Customer
Theresa Swint	2164.0	Normal Customer
Lena Hernandez	2295.0	Normal Customer
Ted Butterfield	1468.0	Normal Customer
Henry MacAllister	736.0	Normal Customer
Bruce Stewart	2562.0	Normal Customer
Andy Yotov	497.0	Normal Customer
Neil Ducich	2893.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Michael Grace	1243.0	Normal Customer
Heather Jas	358.0	Normal Customer
Stefanie Holloman	2096.0	Normal Customer
Kean Takahito	3057.0	Normal Customer
Brenda Bowman	6766.0	AVG Customer
Roland Murray	98.0	Normal Customer
David Bremer	2973.0	Normal Customer
Emily Phan	5478.0	AVG Customer
Justin Hirsh	2256.0	Normal Customer
Robert Marley	5979.0	AVG Customer
Patricia Hirasaki	730.0	Normal Customer
Eudokia Martin	1041.0	Normal Customer
Kean Nguyen	2172.0	Normal Customer
Mary Zewe	2565.0	Normal Customer
Ralph Arnett	2618.0	Normal Customer
Tony Sayre	2243.0	Normal Customer
Sung Chung	281.0	Normal Customer
Parhena Norris	1467.0	Normal Customer
Fred McMath	2523.0	Normal Customer
Jim Radford	1157.0	Normal Customer
Sam Zeldin	3625.0	Normal Customer
Craig Leslie	2354.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Carol Adams	3790.0	Normal Customer
Elpida Rittenbach	1246.0	Normal Customer
Marc Harrigan	2394.0	Normal Customer
Gary Mitchum	3079.0	Normal Customer
Vivek Grady	427.0	Normal Customer
Thais Sissman	5.0	Normal Customer
Tom Boeckenhauer	9134.0	AVG Customer
Rob Haberlin	2086.0	Normal Customer
Tamara Willingham	5279.0	AVG Customer
Adam Shillingsburg	3255.0	Normal Customer
Ken Lonsdale	14175.0	VIP Customer
Michael Granlund	776.0	Normal Customer
Thea Hudgings	2943.0	Normal Customer
Lela Donovan	5.0	Normal Customer
Bill Donatelli	5719.0	AVG Customer
Cari Sayre	1279.0	Normal Customer
Khloe Miller	454.0	Normal Customer
David Philippe	1059.0	Normal Customer
Ruben Dartt	3134.0	Normal Customer
Joy Smith	2516.0	Normal Customer
Cari Schnelling	538.0	Normal Customer
Harold Ryan	5249.0	AVG Customer

Customer_Name	Total_Sales	Customer_Level
Paul Lucas	239.0	Normal Customer
Sharelle Roach	3233.0	Normal Customer
Bart Folk	273.0	Normal Customer
Sylvia Foulston	3337.0	Normal Customer
Stuart Van	1271.0	Normal Customer
Speros Goranitis	3125.0	Normal Customer
Erin Smith	4658.0	Normal Customer
Jas O'Carroll	1934.0	Normal Customer
Rob Dowd	2913.0	Normal Customer
Gary Zandusky	4355.0	Normal Customer
Russell Applegate	4794.0	Normal Customer
Cynthia Arntzen	3042.0	Normal Customer
Georgia Rosenberg	1284.0	Normal Customer
Kelly Lampkin	5016.0	AVG Customer
Victor Preis	994.0	Normal Customer
Joseph Airdo	6491.0	AVG Customer
Jim Mitchum	865.0	Normal Customer
Giulietta Baptist	4716.0	Normal Customer
Alan Haines	1587.0	Normal Customer
Alan Dominguez	6107.0	AVG Customer
Bryan Davis	658.0	Normal Customer
Justin Ellison	2697.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Eric Hoffmann	684.0	Normal Customer
Dorothy Wardle	1205.0	Normal Customer
Todd Sumrall	11892.0	VIP Customer
Christina DeMoss	1206.0	Normal Customer
Evan Minnotte	367.0	Normal Customer
Anna Gayman	3489.0	Normal Customer
William Brown	6160.0	AVG Customer
Jack O'Briant	3255.0	Normal Customer
Tim Taslimi	440.0	Normal Customer
Maria Bertelson	1549.0	Normal Customer
Dean Braden	2333.0	Normal Customer
Sally Matthias	244.0	Normal Customer
Don Weiss	344.0	Normal Customer
Edward Nazzal	2199.0	Normal Customer
Victoria Pisteka	3361.0	Normal Customer
Tiffany House	1022.0	Normal Customer
Sara Luxemburg	3053.0	Normal Customer
Dan Reichenbach	6528.0	AVG Customer
Christopher Martinez	8954.0	AVG Customer
Emily Grady	1832.0	Normal Customer
Bryan Spruell	949.0	Normal Customer
Cathy Prescott	5402.0	AVG Customer

Customer_Name	Total_Sales	Customer_Level
Craig Molinari	3984.0	Normal Customer
Vivek Gonzalez	846.0	Normal Customer
Erica Bern	1643.0	Normal Customer
Lisa DeCherney	1962.0	Normal Customer
Ken Dana	2244.0	Normal Customer
Mark Packer	3206.0	Normal Customer
Bryan Mills	1339.0	Normal Customer
Peter Fuller	9063.0	AVG Customer
Nicole Brennan	274.0	Normal Customer
Candace McMahon	1674.0	Normal Customer
Alex Grayson	661.0	Normal Customer
Dennis Bolton	1081.0	Normal Customer
Susan MacKendrick	1043.0	Normal Customer
Michael Kennedy	2302.0	Normal Customer
Daniel Lacy	3272.0	Normal Customer
Noel Staavos	2965.0	Normal Customer
Mick Hernandez	5503.0	AVG Customer
Ben Peterman	3676.0	Normal Customer
Yoseph Carroll	5454.0	AVG Customer
Liz Willingham	1259.0	Normal Customer
Bobby Odegard	131.0	Normal Customer
Sally Hughsby	3407.0	Normal Customer

Customer_Name	Total_Sales	Customer_Level
Maureen Fritzler	937.0	Normal Customer

In [49]:
```
%%sql

Select Customer_Level, Count(Customer_Level) as Customer_Count
from Customer_Levels
Group By Customer_Level;
```

```
 * mssql+pyodbc://@DESKTOP-CFQC9I1/SuperStores?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.
```

Out[49]:

Customer_Level	Customer_Count
Normal Customer	676
VIP Customer	20
AVG Customer	97

In [ ]:
```

```